

Deep Hole drilling machine (Horizontal type)

machine designed for this purpose is known as deep hole drilling machine. High cutting speeds and less feed are necessary to drill deep holes. A non-rotating drill is fed slowly into the rotating work at high speeds. Coolant should be used while drilling. There are two different types of deep hole drilling machines.

1. Vertical type
2. Horizontal type



2.12 The size of a Drilling Machine

Drilling Machines are specified according to their type.

1. A portable drilling machine is specified by the maximum diameter of the drill that it can hold.
2. The size of the sensitive and upright drilling machines are specified by the size of the largest work piece that can be centered under the spindle. It is slightly smaller than twice the distance between the face of the column and the axis of the spindle.
3. The size of a radial drilling machine is specified by the diameter of the column and length of the arm, spindle speeds and feeds etc.
4. Gang and multi spindle drilling machines is specified by the number of spindles. Length of the work table and type of feed given to the work.
5. Generally a drilling machine is specified by the maximum diameter of the hole that it can drill in the machine, diameter of table, maximum travel of the spindle, numbers and range of spindle speeds and feeds available.
6. Morse taper number of the drill spindle, floor space required weight of the machine and horse power of the motor.

2.13 Drill spindle Assembly

A drill spindle assembly is illustrated shown in the given figure.

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Drill spindle assembly

The spindle is a vertical shaft which holds the drill. A long keyway is cut on the spindle and a sliding key connects it with a bevel gear or a stepped cone pulley. It receives motion from the driving motor. The spindle and the sleeve are connected by a thrust bearing.

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